Infant formula (ready-to-feed, liquid): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified standard for ready-to-feed liquid infant formula, stated as-consumed. Every limit is the strictest maximum level any credible government sets for the liquid product form, converted to the as-consumed basis; where no government regulates the exact product, the limit is read across from the nearest government value.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

Variants on this standard: cow and soy. Where a cell differs between them the limit column shows both (value per variant); the derivation below is stated for cow.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	10	government maximum level
Cadmium (Cd)	Tier 1	5 cow / 10 soy	government maximum level
Total arsenic (tAs)	Tier 2	50	human-grade read-across (India)
Inorganic arsenic (iAs)	Tier 1	10	government maximum level

Mercury, total (tHg)	Tier 1	4	government maximum level
Methylmercury (MeHg)	Tier 1	—	not compiled
Nickel (Ni)	Tier 2	100	government maximum level
Tin (Sn)	Tier 2	5,000	government maximum level
Aluminium (Al)	Tier 2	500 cow / 1,000 soy	government maximum level
Total chromium (Cr)	Tier 2	300	human-grade read-across (China)
Hexavalent chromium (Cr-VI)	Tier 2	—	not compiled

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	10	government maximum level
Cadmium (Cd)	5	government maximum level
Total arsenic (tAs)	50	India — human-food analog
Inorganic arsenic (iAs)	10	government maximum level
Mercury, total (tHg)	4	government maximum level
Methylmercury (MeHg)	—	not compiled
Nickel (Ni)	100	government maximum level
Tin (Sn)	5,000	government maximum level
Aluminium (Al)	500	government maximum level
Total chromium (Cr)	300	China — human-food analog
Hexavalent chromium (Cr-VI)	—	not compiled

4 | Cite this standard

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References

[1] China — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] India — human-food maximum levels
Binds tAs. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/infant--formula-ready-to-feed. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

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